

Question 1

In a certain tumor genome-wide study, the researchers divided the genome into many windows of equal length and counted the number of somatic SNVs (single nucleotide variations) in each window. It is known that in a certain type of sample, mutations occur approximately independently and randomly on the genome, and the number of mutations in each 20 kb window follows a Poisson distribution.

It is known that the average mutation rate of this tumor sample is 120 mutations per Mb.

Let the random variable X represent the number of mutations in a randomly selected 20 kb window.

- (1) Calculate the probability of exactly four mutations occurring within this 20 kb window.
- (2) Calculate the probability that there are fewer than 3 mutations in this 20 kb window.
- (3) The researchers defined the "mutation enrichment window" as: a 20 kb window with more than 5 mutations. Calculate the probability that any randomly selected 20 kb window becomes a "mutation enrichment window".

Question 2

2. In a biological tracer experiment, a radioactive isotope is used to label molecules inside cells. Let X denote the lifetime of a radioactive atom before it decays.

Suppose X follows an exponential distribution with rate parameter $\lambda = 0.2 \text{ min}^{-1}$.

Answer the following questions:

- 1) What is the half-life $t_{1/2}$, defined as the time by which 50% of the radioactive atoms have decayed?
- 2) What is the probability that a randomly chosen radioactive atom remains undecayed for more than 10 minutes?
- 3) Suppose an atom has already remained undecayed for 5 minutes. For such an atom, is the probability that it will remain undecayed for more than 10 additional minutes from that moment larger than, smaller than, or unchanged compared with the probability at time 0?

Question 3

Question 3 is about normal distribution.

- (1) What is normal distribution? What is its probability density function?
- (2) For a random variable following a normal distribution $N(\mu, \sigma^2)$, how does increasing or decreasing μ transform the graph of the distribution? How does increasing or decreasing σ affect the graph?
- (3) What is the standard normal distribution? In the standardization transformation, how is the z-score defined?
- (4) Suppose the weights of a batch of eggs follow a normal distribution

$$X \sim N(50 \text{ g}, 5^2).$$

- What is the z-score of an egg that weighs 45g?
- (5) What is the probability that one egg weighs less than 42.3g?
 - (6) Suppose I randomly select one egg from the eggs weighing at least 50g. What is the probability that the selected egg weighs at least 55g?

Question 4:

In single-cell RNA sequencing (scRNA-seq) experiments, researchers often face a trade-off between sample size (number of cells sequenced) and experimental cost. Sample size directly affects the accuracy and stability of gene expression estimates. A lab is studying the expression levels of a marker gene in immune cells. Assuming: the gene expression level approximately follows a normal distribution with mean $\mu = 25$ TPM and standard deviation $\sigma = 8$ TPM.

Three team members use different sequencing strategies: Mark sequences 6 cells per sample, Marina sequences 300 cells per sample, and Piccolo sequences 2000 cells per sample. Each member repeats their sampling process 500 times (with replacement) to evaluate the stability of their estimates. **Using R**, simulate the **sampling** processes for the three members from the normal population. Plot the distribution of their sample means (histogram and fit line), calculate the standard deviation of the **sample means** obtained from the sampling, and compare it with the theoretical value of the standard error.

Hint: Use `rnorm(n, mean, sd)` and `hist()` functions.

Question 5

An agricultural research institute is developing a new compound fertilizer aimed at increasing corn yield. Based on extensive long-term studies and field trial data of similar fertilizers, the population standard deviation of the increase in corn yield per acre caused by such fertilizers is known to be $\sigma = 18.2$ kg/acre.

To evaluate the actual effect of this new fertilizer, researchers applied it to 50 experimental plots. After one growing season, the increase in corn yield for these 50 plots was measured. The sample mean increase in corn yield per mu was found to be $\bar{x} = 75.6$ kg/acre.

Based on this data, calculate the 95% confidence interval for the population mean increase in corn yield caused by this new fertilizer.